

STOCKTON, CA, USA

WMO: 724920

Lat: 37.889N

Lon: 121.226W

Elev: 26

StdP: 14.68

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23237

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 30.7	(c) 33.1	(d) 21.8	(e) 16.4	(f) 51.0	(g) 25.3	(h) 19.3	(i) 48.8	(j) 26.6	(k) 54.6	(l) 24.1	(m) 53.3	(n) 2.9	(o) 130	(p) 0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.8	101.5	69.7	98.3	68.8	95.3	68.0	73.1	96.0	70.8	94.2	69.1	91.9	10.0	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	91.0	84.5	61.7	82.2	80.0	59.9	77.2	77.7	36.6	96.0	34.5	94.5	33.1	92.0	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.6	(b) 19.3	(c) 17.2	(e) 26.8	(f) 107.0	(g) 3.2	(h) 2.5	(i) 24.5	(j) 108.8	(k) 22.6	(l) 110.3	(m) 20.8	(n) 111.7	(o) 18.4	(p) 113.5		
(a) 22.6	(b) 19.3	(c) 17.2	(e) 25.5	(f) 76.4	(g) 3.5	(h) 5.5	(i) 23.0	(j) 80.4	(k) 20.9	(l) 83.6	(m) 19.0	(n) 86.7	(o) 16.5	(p) 90.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.4	47.7	51.3	55.7	60.0	66.7	73.4	77.4	76.4	73.0	64.5	54.2	47.5
	DBStd	11.90	4.90	4.55	5.13	5.89	6.07	6.02	5.13	4.63	5.07	5.15	5.45	5.28
	HDD50	286	102	36	10	2	0	0	0	0	0	0	22	114
	HDD65	2377	535	385	288	170	50	5	0	0	3	71	326	544
	CDD50	4802	32	71	188	302	518	701	850	820	690	449	146	35
	CDD65	1419	0	0	1	20	103	256	385	355	244	54	1	0
	CDH74	19026	0	1	77	459	1586	3474	5061	4452	3008	871	37	0
	CDH80	9999	0	0	6	140	703	1881	2955	2497	1532	284	1	0
Wind	WSAvg	7.5	6.1	7.0	7.3	8.6	9.5	9.6	8.5	8.0	7.2	6.1	5.4	6.4
Precipitation	PrecAvg	14.0	2.8	2.4	2.0	1.0	0.4	0.1	0.0	0.0	0.3	0.7	2.0	2.0
	PrecMax	26.7	7.0	8.2	6.5	3.1	3.7	0.7	0.6	0.8	2.5	2.7	6.2	5.3
	PrecMin	5.6	0.2	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	5.4	2.0	1.9	1.5	0.8	0.7	0.1	0.1	0.2	0.6	0.7	1.6	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.1	72.2	80.2	90.3	98.2	104.0	105.5	103.8	100.4	92.7	78.2	65.2
		MCWB	54.9	58.0	62.0	65.0	68.3	70.7	71.9	70.6	69.1	65.5	60.0	56.4
	2%	DB	62.4	67.7	76.0	83.9	92.4	99.5	101.5	99.8	96.6	86.8	73.9	61.9
		MCWB	54.1	56.5	59.9	62.4	66.0	69.3	70.0	69.2	68.4	63.4	58.6	54.4
	5%	DB	59.3	64.3	72.5	79.5	87.9	95.1	97.8	96.4	93.1	82.7	69.5	59.2
		MCWB	52.9	54.5	58.0	60.3	63.7	67.1	69.2	68.2	66.8	61.1	57.1	53.3
	10%	DB	56.6	61.1	68.5	74.9	83.0	90.7	94.4	92.7	89.3	78.8	65.4	56.8
		MCWB	51.6	53.5	56.5	58.5	62.0	65.7	68.6	67.3	65.4	60.1	54.9	51.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.1	61.0	64.4	67.3	71.7	73.6	78.8	75.2	78.4	67.8	63.3	59.8
		MCDB	62.6	67.5	76.5	83.2	90.7	100.2	95.9	97.1	92.3	86.9	74.0	63.3
	2%	WB	56.1	58.6	61.7	64.1	67.6	70.7	73.0	71.5	70.7	65.2	60.5	57.1
		MCDB	59.5	64.9	73.1	80.4	89.3	95.3	96.5	95.7	90.4	83.0	70.1	59.9
	5%	WB	54.0	56.5	59.5	61.8	65.2	68.7	70.6	69.4	68.0	63.3	58.6	54.8
		MCDB	57.3	62.3	69.6	76.9	85.4	91.7	94.7	92.4	88.5	79.1	66.5	57.3
	10%	WB	52.3	54.5	57.4	59.5	63.0	66.8	68.9	67.9	66.2	61.4	56.8	52.6
		MCDB	55.6	59.8	66.5	72.2	80.8	88.4	92.4	90.2	86.7	75.6	63.6	55.6
Mean Daily Temperature Range	5% DB	MDBR	16.1	19.4	22.7	25.1	28.0	31.1	32.8	32.2	30.9	27.6	21.7	16.9
		MCDBR	21.3	25.1	29.3	33.7	35.8	37.1	36.5	36.1	36.3	34.0	28.0	20.0
		MCWBR	13.5	14.7	15.3	15.1	14.5	14.0	13.8	13.6	14.4	14.8	15.0	12.7
	5% WB	MCDBR	15.3	21.1	26.2	30.5	33.7	34.6	34.6	33.7	32.7	30.1	23.3	12.8
		MCWBR	10.5	13.2	14.6	15.0	15.4	14.1	15.1	13.9	14.7	14.2	13.3	8.9
Clear-Sky Solar Irradiance	taub	0.333	0.331	0.343	0.362	0.365	0.362	0.368	0.352	0.355	0.353	0.342	0.332	
	taud	2.469	2.486	2.439	2.380	2.391	2.411	2.402	2.431	2.425	2.439	2.445	2.478	
	Ebn at Noon	268	285	292	291	290	290	287	290	284	273	262	260	
	Edh at Noon	26	29	33	37	38	37	37	35	33	30	26	24	
All-Sky Solar Radiation	RadAvg	729	1051	1499	1953	2355	2636	2601	2342	1910	1368	896	656	
	RadStd	117	129	150	170	137	98	70	64	62	84	57	101	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-1.12	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)	+0.72	N/A	-2.02	N/A	N/A	N/A	N/A	-211	+239	+54

Nomenclature: See separate page